Test 3

Added 7 pts to everyone's raw score before curving (Q3b, 4)

After adding 7:

A: 68-77
B: 58-67
C: 49-57
D: 40-48

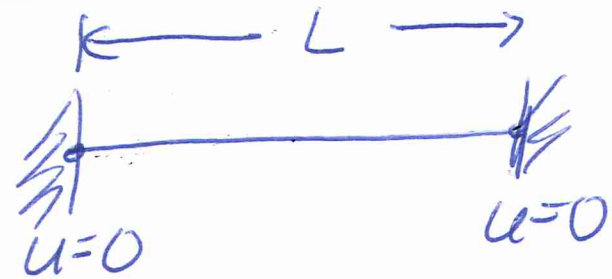
A: ~~/~~
B: ~~/~~

Avg. 72% (after curve)
High 94%

Q3b + 4 are corrected on take home, so for take home there will be no points added.

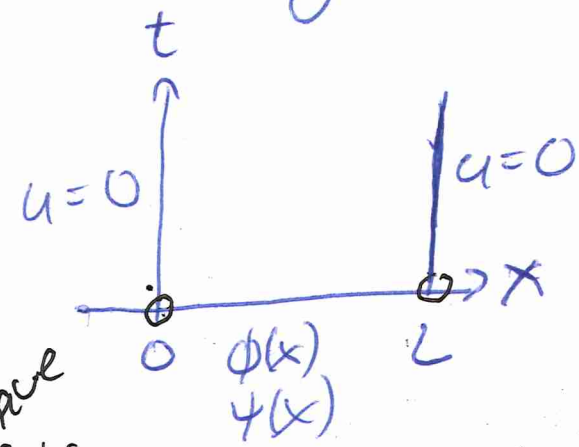
Ch 8 S8 Vibrating String

Going from one BC to two



guitar string.

$$u_{xx} - u_{tt} = 0 \qquad 0<x<L,\ t>0$$
$$u(x,0) = \phi(x) \qquad 0<x<L$$
$$u_t(x,0) = \psi(x) \qquad 0<x<L$$
$$u(0,t) = u(L,t) = 0 \qquad t>0$$



don't have to have for wave eq

$$\begin{cases} \phi(x=0,0) = u(x=0,t=0) = 0 \\ \phi(x=L,0) = u(x=L,t=0) = 0 \end{cases}$$

compatibility condition

Sol'n separation of Variables

1st consider $u_{xx} - u_{tt} = 0$

$u(0,t) = u(L,t) = 0$

Assume $u(x,t) = X(x)\,T(t)$

subst. in

$$\left[X''T - XT'' = 0 \right] \frac{1}{XT}$$

separate

$$\frac{X''}{X} - \frac{T''}{T} = 0 \Rightarrow \frac{X''}{X} = \frac{T''}{T} = -\lambda$$

Space $X'' + \lambda X = 0 \quad 0 < x < L$

BC $u(0,t) = X(0)\,T(t) = 0$

$X(0) = 0 \quad X(L) = 0$

$ab = 0$
$\Rightarrow a = 0$ or $b = 0$

$ab = c$

Case 1 $\lambda < 0$ assume e^{rx}

$$X(x) = c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x}$$

$$\left.\begin{aligned} X(0) &= c_1 + c_2 = 0 \\ X(L) &= c_1 e^{\sqrt{-\lambda}L} + c_2 e^{-\sqrt{-\lambda}L} = 0 \end{aligned}\right\} \quad c_1 = c_2 = 0$$

$$A\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\det A \neq 0$

$X(x) = 0 \Rightarrow u(x,t) = 0$

trivial soln : uninteresting

<u>Case 2</u> $\lambda = 0$ $X'' = 0$

$X'(x) = c_1$ $X(x) = c_1 x + c_2$

$X(0) = c_2 = 0$ $X(x) = c_1 x$ $X(L) = c_1 L = 0$

$\Rightarrow c_1 = 0$

$X(x) = 0$ trivial soln :(

Case 3 $\lambda > 0$

$$X'' + \lambda X = 0$$

$$X(x) = c_1 \cos(\sqrt{\lambda}\, x) + c_2 \sin(\sqrt{\lambda}\, x)$$

$$X(0) = c_1 (1) + c_2 (0) = c_1 = 0$$

$$X(x) = c_2 \sin(\sqrt{\lambda}\, x)$$

$$X(L) = c_2 \sin(\sqrt{\lambda}\, L) = 0$$

$$\sqrt{\lambda}\, L = k\pi \qquad k = 0, \pm 1, \pm 2, \ldots$$

$$\lambda_k = \frac{k^2 \pi^2}{L^2} \qquad k = 1, 2, 3, \ldots$$

<u>eigenvalues</u> for this problem

$$X(x) = \sin\left(\frac{k\pi}{L} x\right)$$

$T(t)$: $T'' + \lambda_k T = 0 \qquad \lambda_k > 0$

$$T_k(t) = A_k \cos\left(\frac{k\pi}{L} t\right) + B_k \sin\left(\frac{k\pi}{L} t\right)$$

No IC here!!

$$u_k(x,t) = X_k(x) T_k(t) = \sin\left(\frac{k\pi x}{L}\right)\left[A_k \cos\left(\frac{k\pi}{L} t\right) + B_k \sin\left(\frac{k\pi}{L} t\right)\right]$$

Now apply I.C.

$$\sum_{k=1}^{\infty} A_k u_k(x,0) = \sum_{k=1}^{\infty} A_k \sin\left(\frac{k\pi x}{L}\right) = \phi(x)$$ Fourier sine series!

$$\sum_{k=1}^{\infty} C_k \frac{\partial u_k}{\partial t}(x,0) = \sum_{k=1}^{\infty} \underbrace{\frac{k}{L}\pi B_k}_{\text{coeff of Fourier sine series of } \psi(x).} \sin\left(\frac{k\pi}{L} x\right) = \psi(x)$$

$$u(x,t) = \sum_{k=1}^{\infty} \sin\left(\frac{k\pi x}{L}\right)\left[A_k \cos\left(\frac{k\pi}{L} t\right) + B_k \sin\left(\frac{k\pi}{L} t\right)\right]$$

$x=0 \quad u=0$ ✓ $\qquad x=L \quad u=0$ ✓

Check

Not going to check ①

Does u satisfy $u_{xx} - u_{tt} = 0$?

① Not responsible for

Yes. We need 2nd deriv. to make sense (series converges absolutely)

$\Rightarrow |A_k| \le M k^{-4} \qquad |B_k| \le M k^{-4}$

Does u satisfy BC? Yes

IC? Need F.S. of $\phi(x)$ to converge to $\phi(x)$

② Needed for well posedness

$\Rightarrow \phi$ is piecewise smooth (~~sectio~~ contin. & sectionally C^1) $[0, L]$

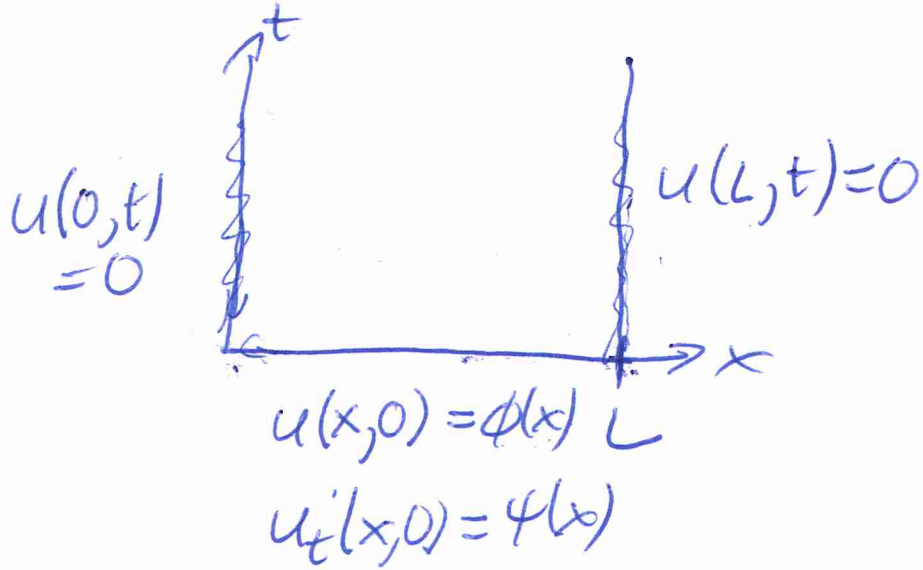
Don't need $\phi(0) = \phi(L)$

$\Rightarrow \psi(x)$ is piecewise smooth.

③ compatibility condition

$\phi(0) = 0$

$\phi(L) = 0$

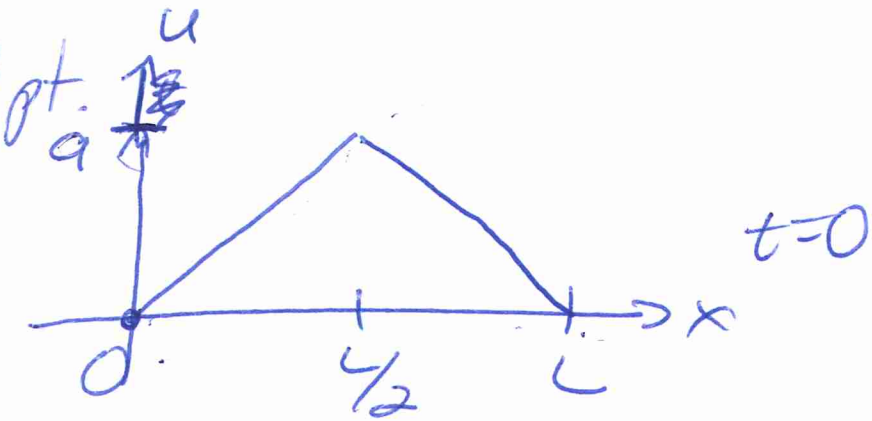


In this class we should check ② & ③

$\Rightarrow$ problem is well-posed.

example 8.1

string + pluck at midpt.



$$\phi(x) = \frac{2a}{L}\begin{cases} x & 0 \le x < \frac{L}{2} \\ L-x & \frac{L}{2} \le x \le L \end{cases}$$

$\psi(x) = 0$

$u(0,t) = u(L,t) = 0$ Guaranteed to be well-posed?

- compatibility ok? Yes
- $\phi(x), \psi(x)$ piecewise smooth? Yes.

Find the F.S. sine series (extend oddly so that $u(x=0,t)=0$).

$\psi(x) = 0 \Rightarrow B_k = 0$

$$A_k \cancel{B_k} = \frac{2}{L}\int_0^L \phi(x)\sin\left(\frac{k\pi x}{L}\right)dx$$

$$= \frac{2}{L}\frac{2a}{L}\int_0^{L/2} x\sin\left(\frac{k\pi x}{L}\right)dx + \frac{2}{L}\frac{2a}{L}\int_{L/2}^{L}(L-x)\sin\left(\frac{k\pi x}{L}\right)dx$$

$\vdots$

$$= \frac{8a}{k^2\pi^2}\sin\left(\frac{k\pi}{2}\right)$$

$$u(x,t) = \frac{8a}{\pi^2} \sum_{k=1}^{\infty} \frac{\sin(k\pi/2)}{k^2} \sin\left(\frac{k\pi}{L}x\right) \overset{\cos}{\cancel{\sin}}\left(\frac{k\pi}{L}t\right)$$

If k is even, then $u(x,t)$ is zero.

so $k = 2n+1$

$$u(x,t) = \frac{8a}{\pi^2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} \sin\left(\frac{(2n+1)\pi x}{L}\right) \overset{\cos}{\cancel{\sin}}\left(\frac{(2n+1)\pi}{L}t\right)$$

$$u_{xx} - \frac{1}{c^2} u_{tt} = 0$$

write above sol'n with t', + then replace t' by ct.

Recall: $$u(x,t) = \sum_{k=1}^{\infty} \sin\left(\frac{k\pi x}{L}\right)\left[\underbrace{A_k}_{\text{assoc. with } \phi(x)} \cos\left(\frac{k\pi t}{L}\right) + \underbrace{B_k}_{\text{assoc. with } \psi(x)} \sin\left(\frac{k\pi t}{L}\right)\right]$$

$$\phi(x) = \sum_{k=1}^{\infty} A_k \sin\left(\frac{k\pi x}{L}\right)$$

$$\psi(x) = \sum_{k=1}^{\infty} \frac{k\pi}{L} B_k \sin\left(\frac{k\pi x}{L}\right) \qquad B_k = \frac{L}{k\pi}\frac{2}{L}\int_0^L \psi(x) \sin\left(\frac{k\pi x}{L}\right)$$

Physical ~~inter~~ problem.

$$c = \sqrt{\frac{T}{\rho}} \qquad T = \text{tension}, \quad \rho = \text{density}$$

Using a trig. identity.

$$u_k(x,t) = \sin\left(\frac{k\pi x}{L}\right)\left[A_k \cos\left(\frac{k\pi c t}{L}\right) + B_k \sin\left(\frac{k\pi c t}{L}\right)\right]$$

$$= E_k \sin\left(\frac{k\pi x}{L}\right) \cos\left[k\omega(t - t_k)\right]$$

↑ max displacement

where $\omega = \frac{\pi}{L}\sqrt{\frac{T}{\rho}}$ → related to frequency

$\omega = 2\pi f$

t_k = shift

ω = angular freq. $[\omega] = \frac{rad}{s}$

↑ term in front of t in sin + cos

i.e. $\cos(\omega t)$

$A_k, B_k \longrightarrow E_k, t_k$ HW.

f = freq. $[f] = \frac{cycles}{s}$ e.g. Hz

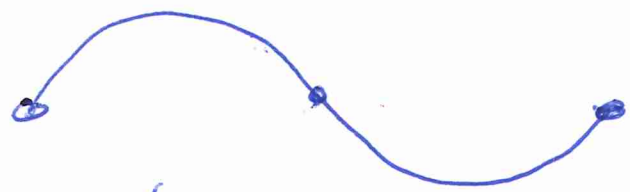
9 Oct 11

(Aside)

Higher modes (k high) produces "timbre"

$k=1$ fundamental mode.

Each mode has fixed pts.



and by ~~fixed~~ fixing a pt.
one can eliminate all modes which does not have that pt as a fixed pt.